



Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations

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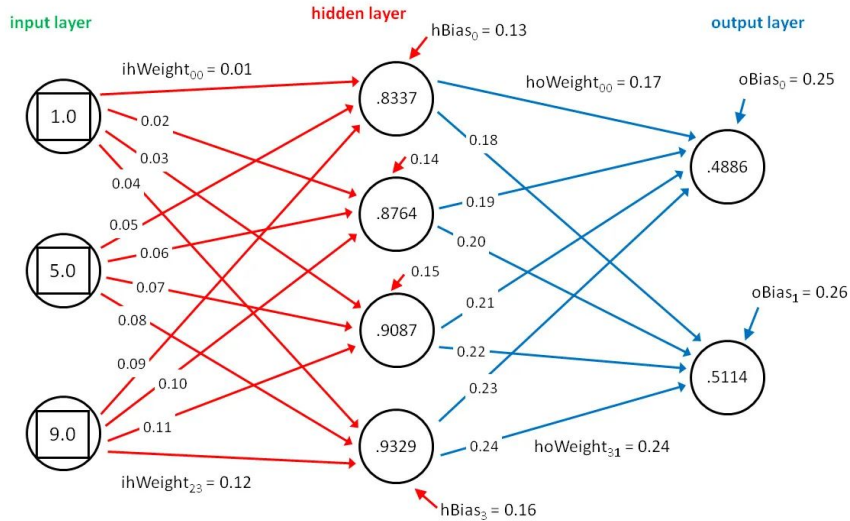
Outline



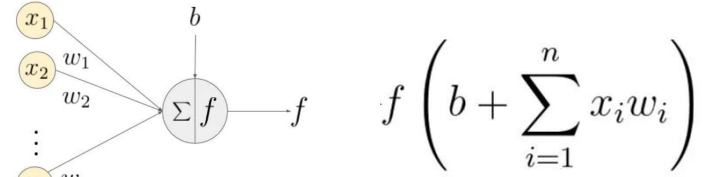
1. Introduction
 - Neural Networks Architecture
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Introduction

Neural Networks Architecture



(ที่มา: <https://visualstudiomagazine.com/articles/2014/11/01/use-python-with-your-neural-networks.aspx>)

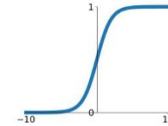


(ที่มา: <https://medium.com/coinmonks/the-mathematics-of-neural-network-60a112dd3e05>)

Activation Functions

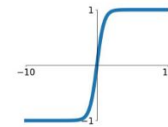
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



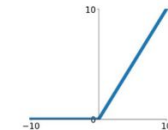
tanh

$$\tanh(x)$$



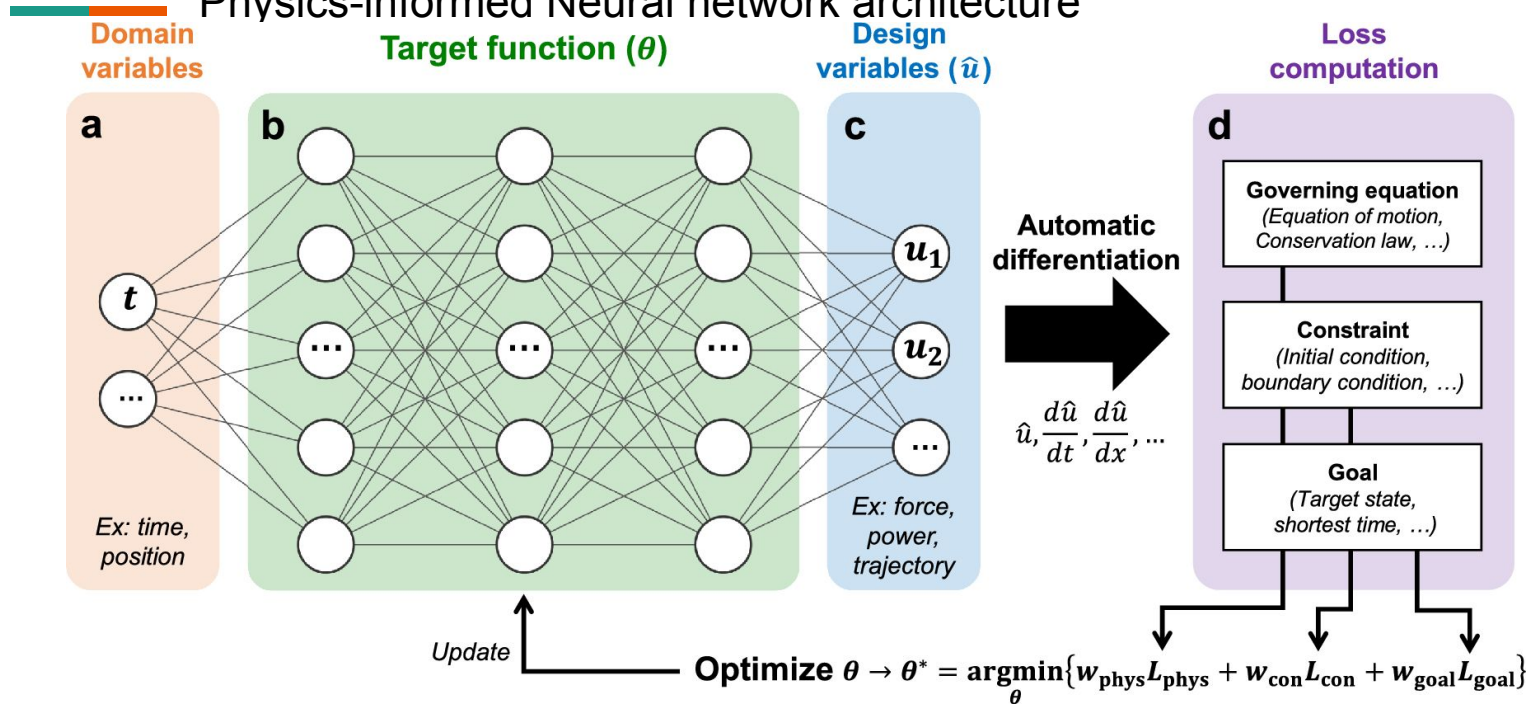
ReLU

$$\max(0, x)$$



Introduction

Physics-informed Neural network architecture



(ที่มารูป: <https://www.nature.com/articles/s41598-023-49977-3>)

Introduction

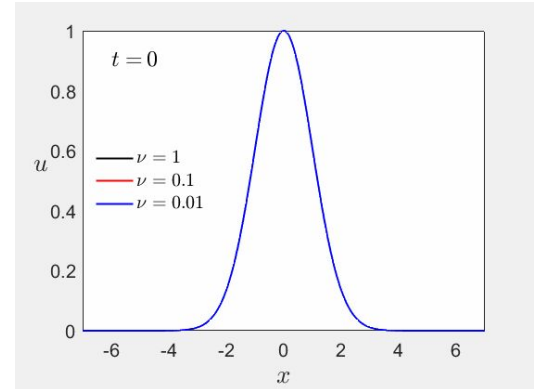
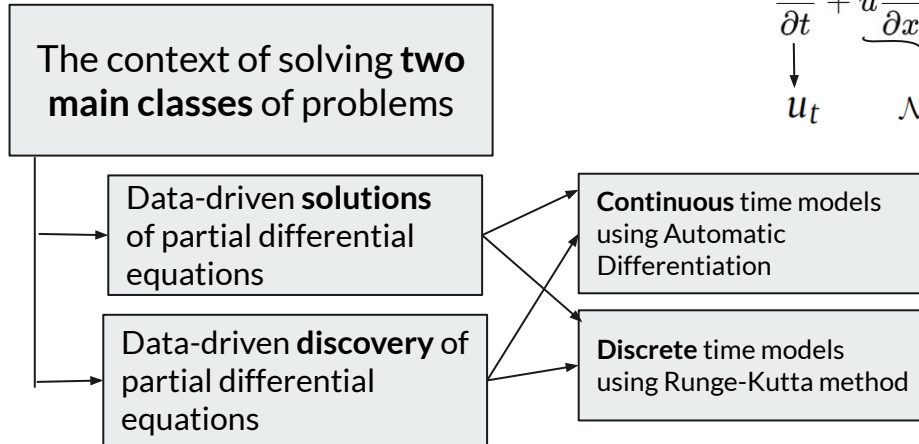
Problem Setup

Parametrized and nonlinear partial differential equations of the general form

$$u_t + \mathcal{N}[u; \lambda] = 0, \quad x \in \Omega, \quad t \in [0, T], \quad (1)$$

$u(t, x)$ denotes the latent (hidden) solution, $\mathcal{N}[u; \lambda]$ is a nonlinear differential operator

$$\underbrace{\frac{\partial u}{\partial t}}_{u_t} + u \underbrace{\frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2}}_{\mathcal{N}[u; \lambda]} = 0 \quad \text{for example: the one 1D the burgers}$$



Results and Discussion

Data-driven solutions of partial differential equations (Continuous time models)

Example for Schrodinger equation

$$ih_t + 0.5h_{xx} + |h|^2h = 0, \quad x \in [-5, 5], \quad t \in [0, \pi/2]. \quad (5)$$

$$h(0, x) = 2 \operatorname{sech}(x), \rightarrow \text{MSE}_0$$

$$\left. \begin{aligned} h(t, -5) &= h(t, 5), \\ h_x(t, -5) &= h_x(t, 5), \end{aligned} \right\} \rightarrow \text{MSE}_b$$

where $\mathbf{h}(\mathbf{t}, \mathbf{x})$ is the **complex-valued** solution.

Physics-informed neural networks $\mathbf{f}(\mathbf{t}, \mathbf{x})$:

$$f := ih_t + 0.5h_{xx} + |h|^2h, \rightarrow \text{MSE}_f$$

Mean Squared Error Loss (**MSE**) for the shared parameters of $h(t, x)$ and $f(t, x)$

$$\text{MSE} = \text{MSE}_0 + \text{MSE}_b + \text{MSE}_f, \quad (6)$$

where

$$\text{MSE}_0 = \frac{1}{N_0} \sum_{i=1}^{N_0} |h(0, x_0^i) - h_0^i|^2,$$

$$\text{MSE}_b = \frac{1}{N_b} \sum_{i=1}^{N_b} \left(|h^i(t_b^i, -5) - h^i(t_b^i, 5)|^2 + |h_x^i(t_b^i, -5) - h_x^i(t_b^i, 5)|^2 \right),$$

and

$$\text{MSE}_f = \frac{1}{N_f} \sum_{i=1}^{N_f} |f(t_f^i, x_f^i)|^2.$$

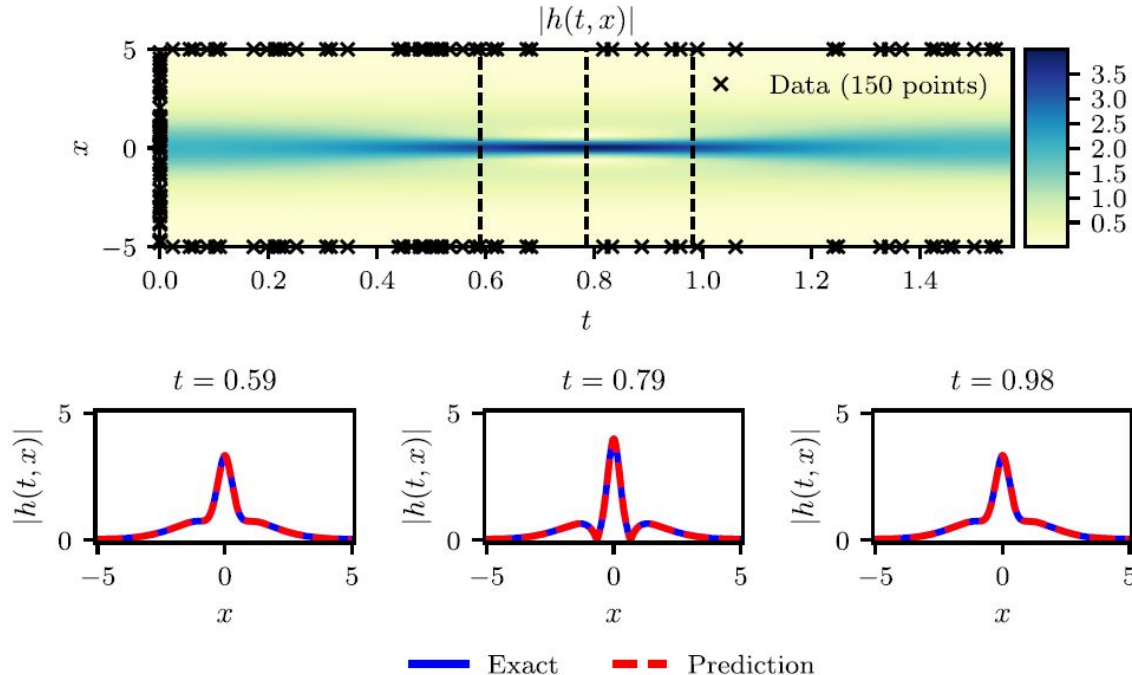
ตัวอย่างผังงานของไฟล์ train.py ของ PINNs partial differential Equations



Result and Discussion

Data-driven solutions of partial differential equations (Continuous time models)

Example for Schrodinger equation



Result and Discussion

Data-driven solutions of partial differential equations (Discrete time models) - Allen–Cahn equation

$$u_t - 0.0001u_{xx} + 5u^3 - 5u = 0, \quad x \in [-1, 1], \quad t \in [0, 1], \quad (12)$$

$$u(0, x) = x^2 \cos(\pi x),$$

$$u(t, -1) = u(t, 1),$$

$$u_x(t, -1) = u_x(t, 1).$$

The nonlinear operator

$$\mathcal{N}[u^{n+c_j}] = -0.0001u_{xx}^{n+c_j} + 5(u^{n+c_j})^3 - 5u^{n+c_j},$$

The general form of Runge–Kutta methods

$$u^{n+c_i} = u^n - \Delta t \sum_{j=1}^q a_{ij} \mathcal{N}[u^{n+c_j}], \quad i = 1, \dots, q,$$

$$u^{n+1} = u^n - \Delta t \sum_{j=1}^q b_j \mathcal{N}[u^{n+c_j}].$$

The sum of squared errors

$$SSE = SSE_n + SSE_b,$$

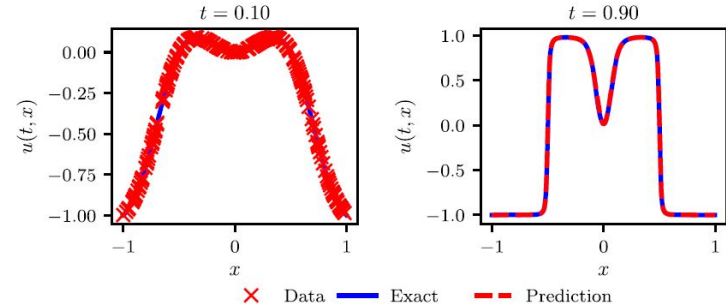
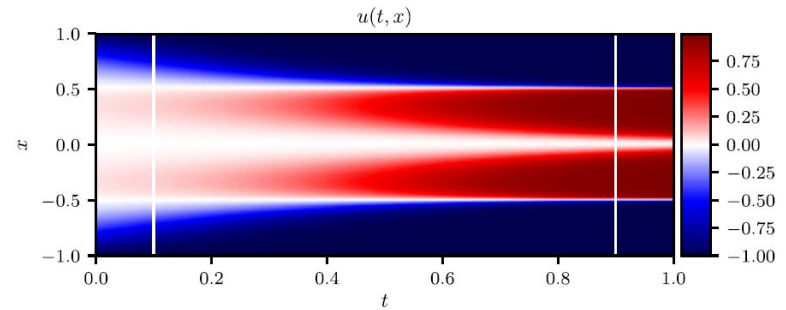
where

$$SSE_n = \sum_{j=1}^{q+1} \sum_{i=1}^{N_n} |u_j^n(x^{n,i}) - u^{n,i}|^2, \quad (13)$$

and

$$SSE_b = \sum_{i=1}^q |u^{n+c_i}(-1) - u^{n+c_i}(1)|^2 + |u^{n+1}(-1) - u^{n+1}(1)|^2 \\ + \sum_{i=1}^q |u_x^{n+c_i}(-1) - u_x^{n+c_i}(1)|^2 + |u_x^{n+1}(-1) - u_x^{n+1}(1)|^2.$$

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Result and Discussion

Data-driven discovery of partial differential equations (Continuous time models) - Navier–Stokes equation

Navier-Stokes equations:

$$\begin{aligned} u_t + \lambda_1(uu_x + vu_y) &= -p_x + \lambda_2(u_{xx} + u_{yy}), \\ v_t + \lambda_1(uv_x + vv_y) &= -p_y + \lambda_2(v_{xx} + v_{yy}), \end{aligned} \quad (15)$$

where $\mathbf{u}(t, x, y)$ denotes the $\mathbf{x}$ -component of the velocity field, $\mathbf{v}(t, x, y)$ the $\mathbf{y}$ -component, $\mathbf{p}(t, x, y)$ the pressure. and, $\boldsymbol{\lambda} = (\lambda_1, \lambda_2)$ are the unknown parameters

Physics-informed neural networks $\mathbf{f}(t, \mathbf{x}, \mathbf{y})$ and $\mathbf{g}(t, \mathbf{x}, \mathbf{y})$:

$$\begin{aligned} f &:= u_t + \lambda_1(uu_x + vu_y) + p_x - \lambda_2(u_{xx} + u_{yy}), \\ g &:= v_t + \lambda_1(uv_x + vv_y) + p_y - \lambda_2(v_{xx} + v_{yy}), \end{aligned} \quad (18)$$

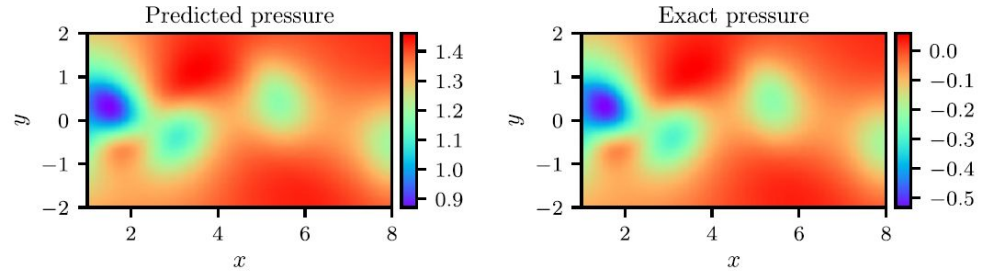
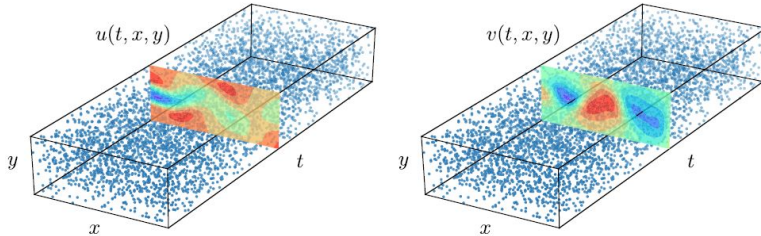
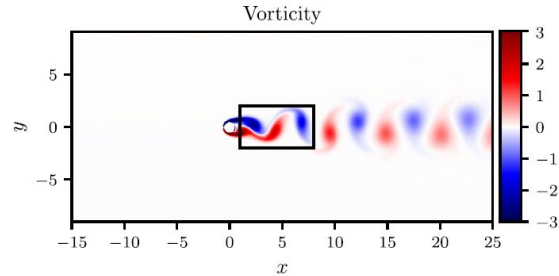
The mean squared error loss:

$$MSE := \frac{1}{N} \sum_{i=1}^N \left(\underset{\substack{\downarrow \\ \text{prediction}}}{|u(t^i, x^i, y^i)|^2} - \underset{\substack{\downarrow \\ \text{data}}}{u^i} \right)^2 + \underset{\substack{\downarrow \\ \text{prediction}}}{|v(t^i, x^i, y^i)|^2} - \underset{\substack{\downarrow \\ \text{data}}}{v^i} \right)^2 + \frac{1}{N} \sum_{i=1}^N \left(\underset{\substack{\downarrow \\ \text{Physics-informed neural networks } \mathbf{f}(t, \mathbf{x}, \mathbf{y})}}{|f(t^i, x^i, y^i)|^2} + |g(t^i, x^i, y^i)|^2 \right). \quad (19)$$

Result and Discussion

Data-driven discovery of partial differential equations (Continuous time models) - Navier–Stokes equation

Incompressible flow and dynamic vortex shedding past a circular cylinder



Correct PDE	$u_t + (uu_x + vu_y) = -p_x + 0.01(u_{xx} + u_{yy})$ $v_t + (uv_x + vv_y) = -p_y + 0.01(v_{xx} + v_{yy})$
Identified PDE (clean data)	$u_t + 0.999(uu_x + vu_y) = -p_x + 0.01047(u_{xx} + u_{yy})$ $v_t + 0.999(uv_x + vv_y) = -p_y + 0.01047(v_{xx} + v_{yy})$
Identified PDE (1% noise)	$u_t + 0.998(uu_x + vu_y) = -p_x + 0.01057(u_{xx} + u_{yy})$ $v_t + 0.998(uv_x + vv_y) = -p_y + 0.01057(v_{xx} + v_{yy})$

We observe that the physics-informed neural network is able to correctly identify the unknown parameters λ_1 and λ_2 with very high accuracy even when the training data was corrupted with noise.

Specifically, for the case of noise-free training data, the error in estimating λ_1 and λ_2 is 0.078%, and 4.67%, respectively. The predictions remain robust even when the training data are corrupted with 1% uncorrelated Gaussian noise, returning an error of 0.17%, and 5.70%, for λ_1 and λ_2 , respectively.

Result and Discussion

Data-driven discovery of partial differential equations (Discrete time models) - Korteweg–de Vries equation

The KdV equation and Initial condition:

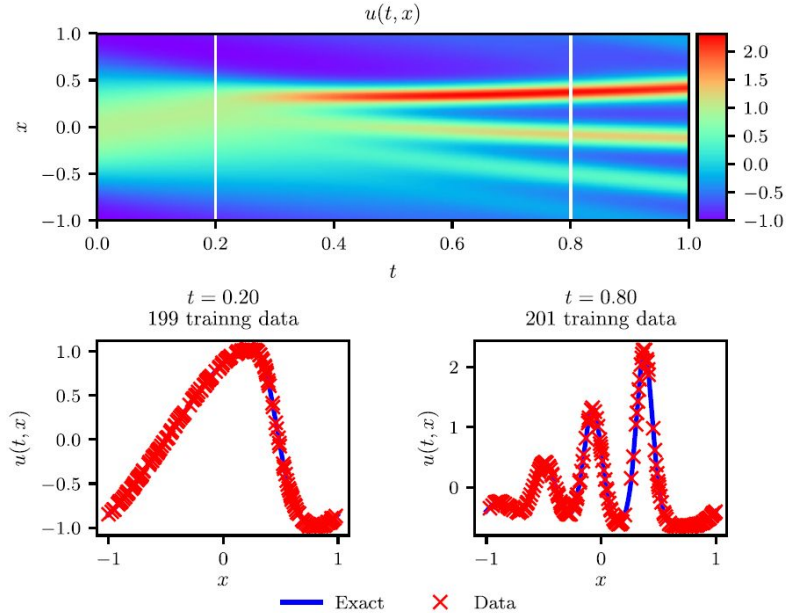
$$u_t + \lambda_1 u u_x + \lambda_2 u_{xxx} = 0,$$

$$u(0, x) = \cos(\pi x)$$

The nonlinear operator:

$$\mathcal{N}[u^{n+c_j}] = \lambda_1 u^{n+c_j} u_x^{n+c_j} - \lambda_2 u_{xxx}^{n+c_j}$$

for the case of noise-free training data, the error in estimating λ_1 and λ_2 is 0.023%, and 0.006%, respectively, while the case with 1% noise in the training data returns errors of 0.057%, and 0.017%, respectively.



Correct PDE	$u_t + uu_x + 0.0025u_{xxx} = 0$
Identified PDE (clean data)	$u_t + 1.000uu_x + 0.0025002u_{xxx} = 0$
Identified PDE (1% noise)	$u_t + 0.999uu_x + 0.0024996u_{xxx} = 0$

Conclusions

- PINNs ที่พัฒนาด้วยไลบรารี Pytorch มีความสามารถทำนายค่าของผลเฉลยได้แม่นยำ
- ค้นพบสมการที่เหมาะสมทางฟิสิกส์ได้อย่างมีความแม่นยำ
- สามารถทำนายข้อมูลทางฟิสิกส์ที่ไม่รู้ค่าได้เช่น ใน ปัญหาแบบฉบับในของไหล, ปฏิกริยา-การแพร่ของระบบ, การแพร่คลื่นน้ำที่ไม่เป็นเชิงเส้น และ กลศาสตร์ควอนตัม



ขอขอบคุณที่เข้าร่วมรับฟัง